COMPARING THEORIES IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES: MORITA EQUIVALENCE AS A JOIN OF COMPLETIONS, THE FORCED ORDER OF COPRODUCTS AND QUOTIENTS, AND CONCEPTUAL COMPLETENESS AS AN IDENTITY OF SAMENESSES

RYOTA SHIBUSAWA

Abstract. Theories are compared through a hierarchy of equivalence relations: definitional equivalence, Morita equivalence (Barrett–Halvorson) and categorical equivalence (equivalent categories of models). We read them as classes of interpretations in the lattice of two-out-of-three classes of the 2-category of coherent categories, a completion defining the class of interpretations it inverts. Morita equivalence is the class inverted by the pretopos completion, the join of the classes inverted by the coproduct completion and by the quotient completion, reached by one alternation with coproducts first; the order is forced, the coherent propositional theory with one atom separating the two orders, and immaterial for classical theories. Hence every Morita equivalence of coherent theories is witnessed by spans of at most four extensions on each side, three for classical theories with a sort of at least two elements. By the Makkai–Reyes conceptual completeness theorem an interpretation inducing an equivalence of model categories is a Morita equivalence: at the level of interpretations the categorical and the Morita classes coincide; hence Hudetz's definable categorical equivalence implies Morita equivalence when the morphisms of models are the elementary embeddings, and Morita bi-interpretability in the sense of Meadows implies Morita equivalence, reversing an arrow left open in his table. The conceptual distance of Khaled, Székely, Lefever and Friend is the alternation depth of a join of two classes, finite at every related pair but unbounded.

1. Introduction

How does one decide that two theories, formulated in different vocabularies, say the same thing? The question has a long history in logic (definitional equivalence, synonymy, bi-interpretability) and a recent one in the philosophy of science, where Barrett and Halvorson [3] proposed *Morita equivalence* as a criterion lying strictly between definitional equivalence and categorical

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Corresponding author. E-mail: r-shibusawa@daiichi-koudai.ac.jp.

equivalence, Tsementzis [22] identified it with the equivalence of pretopos completions, Hudetz [7] proposed definable categorical equivalence, McEldowney [17] and Meadows [18] mapped the relations between the criteria, and Khaled, Székely, Lefever and Friend [10] measured the distance between theories by the number of concepts separating them. The present paper reads these criteria inside one algebraic object, the *sameness lattice* $\mathrm{Same}(\mathcal{C})$ of the category $\mathcal{C}$ of theories and interpretations [19]: the set of classes of interpretations containing the isomorphisms, closed under composition and satisfying two-out-of-three, ordered by inclusion. Each criterion of equivalence is a point of this lattice, namely the class of interpretations which it declares to be identifications; comparing criteria is comparing points; and the lattice operations ask new questions, in particular about joins, which our earlier work [20, 21] has shown to be governed by the finiteness of an alternation.

What is new. We work in the 2-category **Coh** of small coherent categories, the syntactic categories of coherent theories (classical first-order theories enter through their Morleyisation). A completion R of coherent categories, a reflector onto a reflective sub-2-category, inverts a class V_R of interpretations, the largest sameness it sends to equivalences [19, Prop. 5.1].

(a) *Morita equivalence is a join of two completions, reached in one round, in a forced order* (Theorem 4.1, Theorem 4.2). The pretopos completion P is the composite EF of the completion F by finite coproducts and the completion E by quotients of equivalence relations, and $V_{\mathsf{P}} = V_{\mathsf{F}} \vee V_{\mathsf{E}}$ in $\mathrm{Same}(\mathbf{Coh})$; V_{P} is Morita equivalence read as a class of interpretations [22]. The other order fails: $\mathsf{FE} \neq \mathsf{P}$ already for the three-element chain $\bot < p < \top$, the syntactic category of the coherent propositional theory with one atom p, where the quotient of $\top + \top$ gluing the two copies along p (a sort with two elements which become equal when p holds) is not a coproduct of quotients. For Boolean coherent categories, i.e. for classical theories, the two completions commute (Theorem 4.4).

(b) *Lengths of Morita chains* (Theorem 5.1). With one Morita extension allowed to add all sorts definable over the current signature, every Morita equivalence of coherent theories is witnessed by spans of at most four extensions on each side, in the order products, subsorts, coproducts, quotients, and at most three for classical theories having a sort with at least two elements, by the coding of coproduct sorts of [17, Prop. 5.11]; products must precede subsorts and coproducts must precede quotients. The count takes product and coproduct sorts of any finite arity in one step, as in [22]; with the binary sorts of [3] no uniform bound exists (Proposition 5.4).

(c) *Categorical equivalence as a class of interpretations is Morita equivalence* (Theorem 6.1). By the Makkai–Reyes conceptual completeness theorem, an interpretation whose induced functor on categories of

models is an equivalence is inverted by the pretopos completion; this is the observation of Halvorson and Tsementzis [6, §1.3, fn. 8] read as the identity $V_{\mathrm{Mod}} = V_{\mathsf{P}}$ of two samenesses. The strict inclusion of Morita equivalence in categorical equivalence found in [3, Thm. 5.2] therefore concerns pairs of theories with equivalent model categories but no interpretation inducing the equivalence; it is not visible in the lattice. As a consequence, Hudetz's definable categorical equivalence implies Morita equivalence without the "external auxiliaries" of [7, Thm. 2] as soon as the morphisms of models are the elementary embeddings (Corollary 6.3), and a Morita interpretation in the sense of Meadows [18] whose functor of model categories is an equivalence is a Morita equivalence; so Morita bi-interpretability implies Morita equivalence, which reverses an arrow left open in the table of [18, §5.1.1] (Corollary 6.4).

(d) *Conceptual distance is an alternation depth* (Proposition 7.1). The step distance of [10] on a cluster network (X, E, S) is the least number of S-steps in an alternating chain, i.e. the alternation depth of the join $E \vee S$; the theorem of [10] that every finite distance occurs is the failure of a uniform bound in a join whose alternation is finite at every pair it relates, the phenomenon studied for closure operators in [21].

Section 8 reads the results as a statement about duality in the sense of Maruyama's programme [16]: a criterion of equivalence is "dualisable" when it is the class inverted by a completion and the completion is recovered from the models; the category of models detects the Morita sameness (Makkai–Reyes) and the ultracategory of models moreover recovers the pretopos (Makkai); the groupoid of models and isomorphisms, without topology, is left as a question.

What is known. The hierarchy definitional $\subsetneq$ Morita $\subsetneq$ categorical is [3]; Meadows [18] arranges twelve criteria in a table with a definitional, a Morita and a categorical column and rows for isomorphism, equivalence, objective equivalence and mutual interpretability of model categories, shows that Morita interpretations induce functors of model categories [18, Thm. 42] and that mixed and pure Morita extensions give the same equivalence [18, Cor. 47], answers two questions of [3], and leaves the reversal of most arrows of the table open; the identification of Morita equivalence with the equivalence of pretopos completions, and the explicit span through the syntactic category, products first, then subsorts, coproducts, quotients, is [22]; that pretoposes are the maximal enlargements of a coherent category with the same semantics is Makkai–Reyes conceptual completeness [15], in the form we use [12, Cor. 12, Rem. 13]; the ultracategory form is [13]. Our contribution is the lattice reading, the join and order theorems, the counting of extensions, and the consequences for [3, 7, 10, 18]. The finite-alternation criterion for joins of classes of closure operators is [21, Thm. 2.3]; the largest

inverted sameness is [19]. That an equivalence of model categories induced by an interpretation is a Morita equivalence is noted in [6, §1.3, fn. 8] as a consequence of conceptual completeness [8, D3.5.9]; Theorem 6.1 is this remark in the lattice, and its consequences for [7] and [18] are drawn here.

2. Samenesses, inverted classes and joins

Definition 2.1 ([19])**.** A *sameness* on a category $\mathcal{C}$ is a class W of morphisms containing the isomorphisms, closed under composition and satisfying two-out-of-three: if two of f, g, gf lie in W, so does the third. $\mathrm{Same}(\mathcal{C})$ is the complete lattice of samenesses under inclusion; meets are intersections. For a functor $F\colon \mathcal{C} \to \mathcal{D}$, $V_F := \{f : F(f) \text{ invertible}\}$ is the largest sameness F inverts [19, Prop. 5.1]. For a 2-category we use the same words with "invertible" read as "an equivalence" and two-out-of-three up to invertible 2-cells; $\mathrm{Same}(\mathcal{C})$ is then the lattice of such classes.

Two theories T, T' are *W-equivalent* if they are connected by a zig-zag of morphisms in W; this is the equivalence relation on objects induced by W. A criterion of equivalence which is not induced by any sameness is invisible to the lattice; we shall meet one in Section 6.

The following lemma is the categorical form of the one-round case of [21, Thm. 2.3(iv)], where it is shown that for closure operators ρ, ρ' on a lattice the join of the classes $W_\rho \vee W_{\rho'}$ is the class of the join $\rho \sqcup \rho'$ iff the alternation $\rho\rho'\rho\rho'\cdots$ stops at a finite stage at every point, and that one step suffices iff $\rho\rho'$ is idempotent.

Lemma 2.2. *Let R_1, R_2 be reflectors of a 2-category $\mathcal{C}$ onto reflective sub-2-categories, with units $\eta_i\colon X \to R_iX$, and suppose that $R := R_2R_1$ is again a reflector (its image is reflective with unit $X \to R_1X \to R_2R_1X$) and that R inverts the units of R_1 and of R_2. Then $V_R = V_{R_1} \vee V_{R_2}$ in $\mathrm{Same}(\mathcal{C})$, and every $f \in V_R$ is a zig-zag of one R_1-unit, one R_2-unit, an equivalence, and the inverses of one R_2-unit and one R_1-unit.*

Proof. $V_{R_i} \subseteq V_R$ since R inverts the units of R_i: if R_if is an equivalence, the square $R_if \circ \eta_i^X = \eta_i^Y \circ f$ and two-out-of-three in V_R give $f \in V_R$. Conversely let $f\colon X \to Y$ with Rf an equivalence. The unit $X \to RX$ is the composite of $\eta_1^X \in V_{R_1}$ and $\eta_2^{R_1X} \in V_{R_2}$, so it lies in the join; likewise for Y; and $Rf \circ (X \to RX) = (Y \to RY) \circ f$ up to an invertible 2-cell. By two-out-of-three f lies in the join. □

3. Coherent categories and their completions

Coh is the 2-category of small coherent categories (finite limits, images, finite unions of subobjects, all stable under pullback) and coherent functors [8, A1.4]. A coherent theory T has a syntactic category $\mathcal{C}_T$, and interpretations $T \to T'$ are the coherent functors $\mathcal{C}_T \to \mathcal{C}_{T'}$ [8, D1.4]; every coherent category is the syntactic category of its internal theory [22, §4], so we identify theories with coherent categories. A coherent category is *positive* if it

has finite coproducts which are disjoint and stable, *effective* if every equivalence relation is a kernel pair, and a *pretopos* if both. Pretoposes form a reflective sub-2-category **Pretop** $\subseteq$ **Coh** with reflector P, the pretopos completion [15, 8.4], [8, A3.3].

We realise the completions inside the sheaf topos. For $\mathcal{C} \in \mathbf{Coh}$ let $\mathrm{Sh}(\mathcal{C})$ be the sheaves for the coherent topology; the Yoneda embedding $\mathcal{C} \to \mathrm{Sh}(\mathcal{C})$ is coherent and full. Inside $\mathrm{Sh}(\mathcal{C})$ let

- $\mathsf{F}\mathcal{C}$ be the full subcategory of finite coproducts of representables,
- $\mathsf{E}\mathcal{C}$ be the full subcategory of quotients of representables by equivalence relations,
- $\mathsf{P}\mathcal{C}$ be the full subcategory of quotients of finite coproducts of representables by equivalence relations, the coherent objects.

$\mathsf{F}\mathcal{C}$ is the free positive coherent category on $\mathcal{C}$, $\mathsf{E}\mathcal{C}$ the free effective one (the exact completion of a regular category [4]), and $\mathsf{P}\mathcal{C}$ the pretopos completion; moreover $\mathsf{EF}\mathcal{C} = \mathsf{P}\mathcal{C}$, i.e. *adding coproducts and then quotients gives the pretopos completion* [15, 8.4], [22, p. 20]; that the exact completion of a positive regular category is positive, so that $\mathsf{EF}\mathcal{C}$ is a pretopos, is [11, Thm. 2.3]. All three are reflectors on **Coh** and each inverts the units of the others, since $\mathrm{Sh}(\mathsf{F}\mathcal{C}) \simeq \mathrm{Sh}(\mathsf{E}\mathcal{C}) \simeq \mathrm{Sh}(\mathsf{P}\mathcal{C}) \simeq \mathrm{Sh}(\mathcal{C})$ [8, D3.3].

Definition 3.1. Two coherent theories are *Morita equivalent* if they are connected by a span of Morita extensions in the sense of [3, 22]: an extension by sorts defined as product, coproduct, subsort or quotient sorts of existing sorts, with the symbols introduced by the definitions, modulo renaming of symbols. By [22, Thm. 4.7, Cor. 4.9] this is equivalent to the equivalence of pretopos completions, hence to the V_{P}-equivalence of the syntactic categories. We call V_{P} the *Morita sameness*, V_{F} the *coproduct sameness* and V_{E} the *quotient sameness*.

4. Morita equivalence as a join; the order of coproducts and quotients

Theorem 4.1. *In* $\mathrm{Same}(\mathbf{Coh})$, $V_{\mathsf{P}} = V_{\mathsf{F}} \vee V_{\mathsf{E}}$*: the Morita sameness is the join of the coproduct sameness and the quotient sameness, and the join is reached by one alternation, coproducts first: every Morita equivalence* $f\colon \mathcal{C} \to \mathcal{D}$ *is a zig-zag through* $\mathcal{C} \to \mathsf{F}\mathcal{C} \to \mathsf{EF}\mathcal{C} \xrightarrow{\simeq} \mathsf{EF}\mathcal{D} \leftarrow \mathsf{F}\mathcal{D} \leftarrow \mathcal{D}$.

Proof. Lemma 2.2 with $R_1 = \mathsf{F}$, $R_2 = \mathsf{E}$, $R = \mathsf{P} = \mathsf{EF}$. □

The lemma applies equally to $R_1 = \mathsf{E}$, $R_2 = \mathsf{F}$ provided FE is the pretopos completion. It is not.

Theorem 4.2 (the order is forced)**.** *Let* $\mathcal{C}$ *be the three-element chain* $\bot < p < \top$*, a distributive lattice, hence a coherent category: the syntactic category of the coherent propositional theory with one atom* p *and no axioms. Then* $\mathsf{E}\mathcal{C} = \mathcal{C}$*,* $\mathsf{FE}\mathcal{C} = \mathsf{F}\mathcal{C}$ *is not effective, and* $\mathsf{FE}\mathcal{C} \subsetneq \mathsf{P}\mathcal{C}$*. Hence* $\mathsf{FE} \neq \mathsf{P}$*, and the alternation of* E *and* F *starting with* E *needs three steps where the alternation starting with* F *needs two.*

Proof. A thin category is effective: a reflexive relation on x is a subobject of $x \times x = x$ containing the identity, so it is x, the kernel pair of the identity; thus $\mathsf{E}\mathcal{C} = \mathcal{C}$. $\mathrm{Sh}(\mathcal{C})$ is the topos of sheaves on the Sierpiński space, equivalently the arrow category $\mathbf{Set}^{\to}$, an object being a restriction map $X(\top) \to X(p)$; the representables are $\top = (1 \to 1)$ and $p = (\emptyset \to 1)$, and $\bot = (\emptyset \to \emptyset)$. Finite coproducts of representables are the objects $(n \hookrightarrow m)$ with injective restriction, $n \leq m$. The object $\top + \top = (2 \to 2)$ carries the equivalence relation R identifying the two sections over p and not over $\top$: as a subobject of $(\top + \top)^2 = (4 \to 4)$ it is $R(\top) = \Delta$, $R(p) =$ everything. Its quotient in $\mathrm{Sh}(\mathcal{C})$ is $(2 \to 1)$, which is coherent, so it lies in $\mathsf{P}\mathcal{C}$, but its restriction map is not injective, so it is not in $\mathsf{F}\mathcal{C}$. Hence $\mathsf{FE}\mathcal{C} = \mathsf{F}\mathcal{C} \subsetneq \mathsf{P}\mathcal{C}$. Moreover $\mathsf{F}\mathcal{C}$ is not effective: otherwise it would be a pretopos containing $\mathcal{C}$, the universal property of $\mathsf{P}\mathcal{C}$ would give a coherent functor $\mathsf{P}\mathcal{C} \to \mathsf{F}\mathcal{C}$ over $\mathcal{C}$, and composing with the inclusion $\mathsf{F}\mathcal{C} \subseteq \mathsf{P}\mathcal{C}$ would give an endofunctor of $\mathsf{P}\mathcal{C}$ over $\mathcal{C}$, necessarily equivalent to the identity, so that the inclusion would be essentially surjective, against $(2 \to 1) \notin \mathsf{F}\mathcal{C}$. The last sentence follows: $\mathsf{EF}\mathcal{C} = \mathsf{P}\mathcal{C}$ in two steps, while $\mathsf{E}\mathcal{C} = \mathcal{C}$, $\mathsf{FE}\mathcal{C} = \mathsf{F}\mathcal{C}$ and only $\mathsf{EFE}\mathcal{C} = \mathsf{P}\mathcal{C}$. □

Remark 4.3. The object $(2 \to 1)$ is a sort with two elements which become equal when p holds: the quotient of the two-element sort $\top + \top$ by the relation "$u \sim v$ iff $u = v$ or p". In a classical theory the same sort is the coproduct $\{p\} + \{\neg p\} + \{\neg p\}$ of subterminals, a coproduct of quotients; coherent logic lacks the complement $\neg p$ and the sort is reached only as a quotient of a coproduct. The same argument shows that $\mathsf{FE} \neq \mathsf{P}$ for every distributive lattice with a chain $\bot < a < x$: in the positive completion the relation (x, a, a, x) on (x, x) is an equivalence relation whose coequaliser is the fold onto (x) with kernel pair (x, x, x, x).

Theorem 4.4 (Boolean case)**.** *If $\mathcal{C}$ is a Boolean coherent category (every subobject has a complement), then $\mathsf{FE}\mathcal{C} = \mathsf{EF}\mathcal{C} = \mathsf{P}\mathcal{C}$: every quotient of a finite coproduct of representables by an equivalence relation is a finite coproduct of quotients of representables.*

Proof. Let $X = X_1 + \cdots + X_n$ with X_i representable and $R \subseteq X \times X$ an equivalence relation in $\mathrm{Sh}(\mathcal{C})$, $R = \coprod_{i,j} R_{ij}$ with $R_{ij} \subseteq X_i \times X_j$ (coproducts are disjoint and stable); the R_{ij} are subobjects of representables in $\mathsf{P}\mathcal{C}$, hence representable, since the completions add no subobjects to old objects [4]. We induct on n; $n = 1$ is the definition of $\mathsf{E}\mathcal{C}$. Let $A_1 \subseteq X_1$ be the union of the images of the projections $R_{1j} \to X_1$, $j \geq 2$: a subobject of the representable X_1 in $\mathrm{Sh}(\mathcal{C})$, hence a subobject in $\mathcal{C}$ (the Yoneda embedding is full and preserves images and finite unions), and let $B_1 = X_1 \setminus A_1$ be its complement, again representable. B_1 is R_{11}-saturated: if $b\, R_{11}\, a$ with $a \in A_1$, then $a\, R_{1j}\, y$ for some $j \geq 2$ and y, and transitivity gives $b\, R_{1j}\, y$, so $b \in A_1$. Hence R restricts to an equivalence relation $R_{11}|_{B_1}$ on B_1 and to an equivalence relation R' on $X' = A_1 + X_2 + \cdots + X_n$, with no relations between B_1 and X', and $X/R = B_1/R_{11} + X'/R'$. On X' every element of A_1 is related to an element of $Y := X_2 + \cdots + X_n$, so the composite $Y \to X' \to X'/R'$

is an epimorphism, every R'-class meeting Y, hence a regular epimorphism in the topos $\mathrm{Sh}(\mathcal{C})$; its kernel pair is $R'' = R' \cap (Y \times Y)$, the restriction of R, so $Y/R'' \cong X'/R'$ by effectiveness of equivalence relations in $\mathrm{Sh}(\mathcal{C})$. By induction X'/R' is a finite coproduct of quotients of representables, and so is X/R. All objects named are representables or their quotients, so the decomposition takes place in $\mathsf{FE}\mathcal{C}$. □

Remark 4.5. The induction uses the complement of A_1 only; so the conclusion holds for every coherent $\mathcal{C}$ in which the images of definable relations between distinct sorts are complemented. In the chain of Theorem 4.2 the subobject $p \subseteq \top$ is such an image and has no complement.

5. Lengths of Morita chains

We now count extensions. Following [22, §2], a *Morita extension* $T \subseteq T^+$ adds to the signature of T new sort symbols, each defined as the product $S_1 \times \cdots \times S_n$ of finitely many existing sorts, the coproduct $S_1 + \cdots + S_n$ of finitely many existing sorts, a subsort of an existing sort cut out by a formula, or the quotient of an existing sort by a formula which T proves to be an equivalence relation, together with the function symbols of the definitions (projections, injections, inclusion, quotient map), all definitions being formulas over the *existing* signature; several sorts may be added at once, but a sort defined from a sort introduced in the same step is not allowed ([3, fn. 8]: one Morita extension cannot put predicates on the new sorts). Products and coproducts of arbitrary finite arity in one step are the convention of [22, §2]; [3] and [18] define product and coproduct sorts of two factors only, and the two conventions give the same Morita equivalence but not the same counts, see Proposition 5.4. As in [22, §2], every theory is assumed to have a sort which it proves inhabited, which is used to form a singleton sort; empty definable sets and empty sorts are otherwise allowed. A *Morita chain of length* k is a sequence of k Morita extensions. Tsementzis's construction [22, §4] gives, for every coherent theory T, a chain $T \subseteq T_1 \subseteq \widehat{T}$ with T_1 adding all finite products of sorts and $\widehat{T}$ adding all subsorts of these cut out by formulas and all provably functional relations between them, so that $\mathcal{C}_{\widehat{T}} \simeq \mathcal{C}_T$ and $\widehat{T}$ is logically equivalent to the internal theory of $\mathcal{C}_T$; and then a chain $T_{\mathcal{C}} \subseteq T_{\mathsf{FC}} \subseteq T_{\mathsf{EFC}}$ adding coproduct sorts and then quotient sorts [22, Prop. 4.2].

Theorem 5.1. *With product and coproduct sorts of arbitrary finite arity:*

(i) *Two coherent theories are Morita equivalent iff, after renaming symbols apart, they are connected by a span of Morita chains of length at most four on each side, with the steps in the order: products, subsorts, coproducts, quotients.*

(ii) *Two of the order constraints cannot be reversed: products before subsorts, since a subsort of a product (a relation symbol $R \subseteq \sigma \times \tau$ not a product of unary definable sets) is not a product of subsorts; and*

coproducts before quotients, since a quotient of a coproduct need not be a coproduct of quotients (Theorem 4.2). Subsorts commute with coproducts (a subobject of $X + Y$ is $A + B$, by extensivity) and with quotients (a subsort of X/E is D/E for a saturated $D \subseteq X$, and D/E is a subsort of $X/(E \cup \Delta_X)$); whether the bound four is attained in every order is part of Question 5.6.

(iii) *If T is a classical theory (Boolean syntactic category) with non-empty sorts and a sort σ with $T \vdash \exists u, v\,(u \neq v)$, then chains of length at most three suffice: products, subsorts, and then coproducts of quotients in one step.*

Proof. (i) If T and T' are Morita equivalent their pretopos completions are equivalent [22, Cor. 4.9]. The chain $T \subseteq T_1 \subseteq \widehat{T} \subseteq T^{(3)} \subseteq T^{(4)}$ of [22] recalled above, with $T^{(3)}$ adding the coproduct sorts of the sorts of $\widehat{T}$ and $T^{(4)}$ the quotient sorts of the sorts of $T^{(3)}$, ends in a theory logically equivalent (up to renaming) to the internal theory of $\mathsf{P}\mathcal{C}_T$: by $\mathsf{EF} = \mathsf{P}$ (Section 3) every object of $\mathsf{P}\mathcal{C}_T$ is a quotient of a finite coproduct of objects of $\mathcal{C}_T$, i.e. of sorts of $\widehat{T}$, and every morphism of $\mathsf{P}\mathcal{C}_T$ is a provably functional relation between such sorts, definable in $T^{(4)}$. Two equivalent pretoposes have internal theories which differ by renaming [22, p. 20]. The converse is the definition.

(ii) In the theory with two sorts σ, τ and a relation symbol R, the sort R is the subsort of the product sort $\sigma \times \tau$ cut out by $R(x, y)$; it is not isomorphic to a product of subsorts of σ and τ in the models where R is not a rectangle. For coproducts and quotients, Theorem 4.2: in the theory with one atom p, the quotient of $\top + \top$ is not a coproduct of quotients, and quotient sorts of base sorts give nothing there.

(iii) By Theorem 4.4 every object of $\mathsf{P}\mathcal{C}_T$ is a finite coproduct $\coprod_i D_i/E_i$ of quotients of definable sets D_i, sorts of $\widehat{T}$ (reached in two steps). It remains to obtain such a coproduct as one quotient of one sort of $\widehat{T}$. Using σ as a tag: $D_1/E_1 + D_2/E_2$ is the quotient of the definable set

$$D = \{(u, v, \bar{x}, \bar{y}) : (u = v \wedge \bar{x} \in D_1) \vee (u \neq v \wedge \bar{y} \in D_2)\}$$

by the equivalence relation identifying $(u, v, \bar{x}, \bar{y})$ with $(u', v', \bar{x}', \bar{y}')$ iff either $u = v$, $u' = v'$ and $\bar{x}\, E_1\, \bar{x}'$, or $u \neq v$, $u' \neq v'$ and $\bar{y}\, E_2\, \bar{y}'$. Since sorts are non-empty the fillers $\bar{y}$ (resp. $\bar{x}$) exist, and since $u \neq v$ is satisfiable both tags occur; so the quotient is $D_1/E_1 + D_2/E_2$ in every model, and D is a subsort of a product of base sorts, a sort of $\widehat{T}$. For m summands use $m - 1$ pairs (u_j, v_j) of variables of sort σ and the mutually exclusive, jointly exhaustive tags $\theta_i = \bigwedge_{j<i}(u_j \neq v_j) \wedge (u_i = v_i)$ for $i < m$ and $\theta_m = \bigwedge_{j<m}(u_j \neq v_j)$: let $D = \{(\bar{u}, \bar{v}, \bar{x}_1, \ldots, \bar{x}_m) : \bigvee_i (\theta_i \wedge \bar{x}_i \in D_i)\}$ and let E identify two tuples iff they carry the same tag θ_i and their i-th blocks are E_i-related; every tag is realised and the unused blocks are filled, so $D/E \cong \coprod_i D_i/E_i$ in every model, in one step and without using the new sorts. All these are definable over the signature of $\widehat{T}$, so one more Morita extension reaches the internal theory of $\mathsf{P}\mathcal{C}_T$. □

Remark 5.2. The coding of coproduct sorts by a sort with two provably distinct elements is not new: McEldowney [17, Prop. 5.11] uses it to show that for such theories Morita equivalence implies bi-interpretability in the model-theoretic sense, and Kamsma [9, Fact 8.6] reports the fact, as an equivalence, from a 2018 thesis of E. Washington and transfers it to positive logic; the two statements are compared after Corollary 6.4. What is added in (iii) is the count: three stages, the coproducts of quotients being one quotient.

Remark 5.3. Meadows [18, Lemma 46, Cor. 47] proves a normal form of a different kind: a chain of *mixed* Morita extensions, which may also add relation and function symbols, is a chain of pure extensions followed by a single interpretation, the added symbols being postponed to the end. Theorem 5.1 uses pure extensions only and orders the sort constructors; the two normal forms combine.

Proposition 5.4 (binary sorts)**.** *If product and coproduct sorts are restricted to two factors, as in* [3, 18]*, there is no uniform bound: for every k there are Morita equivalent theories not connected by any span of Morita chains of length k on each side.*

Proof. Let T be the coherent theory with one sort σ, two constants c, d and the axiom $c = d \vdash \bot$, and let T' be T with a product sort $\tau = \sigma^m$, $m > 2^k$, reached from T by iterated binary products; T and T' are Morita equivalent. Suppose chains $T \subseteq T^{(k)}$ and $T' \subseteq T'^{(k)}$ of length k end in definitionally equivalent theories. Let M be the model of T with $|M_\sigma| = 2$ and $M^{(k)}$ its expansion to $T^{(k)}$: in a binary step the cardinality of a new sort is at most the square of the largest cardinality of an existing sort (products $ab \leq \max(a,b)^2$, coproducts $a + b \leq \max(a,b)^2$ for $\max(a,b) \geq 2$, subsorts and quotients do not increase cardinality), so every sort of $M^{(k)}$ has at most 2^{2^k} elements. Definitionally equivalent theories have the same sorts up to renaming and a definitional equivalence does not change the domains, so $M^{(k)}$ is a model $M'^{(k)}$ of $T'^{(k)}$ whose reduct M' is a model of T'; the sort σ of T' need not correspond to the sort σ of T, but $|M'_\sigma| \geq 2$ by the axiom $c = d \vdash \bot$, so the sort τ of $M'^{(k)}$ has $|M'_\sigma|^m \geq 2^m > 2^{2^k}$ elements, a contradiction. □

Remark 5.5 (McEldowney's example)**.** McEldowney [17, p. 20] states, without proof, that the empty theory in one sort s and its extension T^ω by binary product sorts s^m for all m are bi-interpretable in the model-theoretic sense but not Morita equivalent with binary sorts, "for the simple reason that there is no finite sequence of Morita extensions" from one to the other. The cardinality argument proves this with the theory T of Proposition 5.4 in place of the empty theory: a span of length k between T and T^ω would make the model with $|M_\sigma| = 2$ a model of T^ω in which every sort has at most 2^{2^k} elements and σ^m has at least 2^m, for every m. In the convention of [22, Def. 2.2], where a Morita extension may add any set of new sorts, T^ω is

a single Morita extension of T and the two theories are Morita equivalent, as they must be, having the same pretopos completion. Thus with binary sorts Morita equivalence is strictly finer than V_{P} once infinitely many sorts are allowed, and the identification of Morita equivalence with V_{P} (Section 4, Theorem 6.1, [22, Thm. 4.7]) is for the convention of [22].

Question 5.6. Is the bound four in Theorem 5.1(i) attained by a single coherent theory, i.e. is there a coherent theory whose pretopos completion is not reached by any chain of three Morita extensions in any order? The candidate is the theory with sorts σ, τ, a relation $R \subseteq \sigma \times \tau$ and a subobject $A \subseteq R$ without complement, and the sort $(R + R)/{\sim_A}$ gluing two copies of R along A; a proof requires an invariant of the sorts obtainable in three steps.

6. Conceptual completeness: the categorical sameness is the Morita sameness

Let $\mathrm{Mod}(\mathcal{C})$ be the category of coherent functors $\mathcal{C} \to \mathbf{Set}$ and natural transformations; for a Boolean $\mathcal{C}$ (the Morleyisation of a classical theory) the natural transformations are the elementary embeddings, the morphisms of models used in [3]. Mod is a 2-functor $\mathbf{Coh}^{op} \to \mathbf{CAT}$, and V_{Mod} is the class of coherent functors f for which $\mathrm{Mod}(f)$ is an equivalence of categories. Barrett and Halvorson call two theories *categorically equivalent* if their categories of models are equivalent, by any equivalence.

Theorem 6.1. $V_{\mathrm{Mod}} = V_{\mathsf{P}}$ *in* $\mathrm{Same}(\mathbf{Coh})$*: an interpretation induces an equivalence of the categories of models iff it is a Morita equivalence.*

Proof. The unit $\mathcal{C} \to \mathsf{P}\mathcal{C}$ induces an equivalence $\mathrm{Mod}(\mathsf{P}\mathcal{C}) \to \mathrm{Mod}(\mathcal{C})$ [12, Rem. 13], so $\mathrm{Mod}(f)$ is an equivalence iff $\mathrm{Mod}(\mathsf{P}f)$ is. If $\mathrm{Mod}(\mathsf{P}f)$ is an equivalence, then $\mathsf{P}f$, a morphism of pretoposes, is an equivalence by the Makkai–Reyes conceptual completeness theorem [15, Ch. 7], [12, Cor. 12]; so $f \in V_{\mathsf{P}}$. Conversely if $\mathsf{P}f$ is an equivalence so is $\mathrm{Mod}(\mathsf{P}f)$. □

Remark 6.2 (the gap of [3] is invisible to the lattice)**.** Theorem 5.2 of [3] exhibits theories T_1 (one sort, countably many unary predicates p_i, the axiom that the sort is a singleton) and T_2 (the same with q_i, and the axioms $q_0 \to q_i$) whose categories of models are both discrete with $2^{\aleph_0}$ objects, hence equivalent, while T_1, T_2 are not Morita equivalent. By Theorem 6.1 no interpretation $T_1 \to T_2$ or $T_2 \to T_1$ induces that equivalence; indeed the interpretation $p_i \mapsto q_i$ induces the inclusion of the models of T_2 into those of T_1, which is not essentially surjective. Categorical equivalence as an object relation is therefore not the equivalence relation induced by any sameness on **Coh** whose restriction to interpretations is V_{Mod}: the relation induced by V_{Mod} is Morita equivalence. In the terms of [12, Rem. 13], pretoposes are the maximal enlargements with the same semantics, and the lattice sees exactly the enlargements.

Corollary 6.3 (definable categorical equivalence)**.** *Let T_1, T_2 be classical first-order theories with disjoint signatures, and take as morphisms of models the elementary embeddings. If T_1 and T_2 are definably categorically equivalent in the sense of* [7, Def. 4]*, i.e. connected by an equivalence of categories of models given by reconstruction functors, then they are Morita equivalent.*

Proof. A reconstruction functor $\mathrm{Mod}(T_1) \to \mathrm{Mod}(T_2)$ is given by a reconstruction manual [7, Def. 5–7], which in the first-order case is an interpretation $\varphi\colon T_2 \to T_1$ (each sort of T_2 is sent to a definable subset of a product of sorts of T_1, each symbol to a definable relation, and every model of T_1 yields a model of T_2, so T_1 proves the translated axioms by completeness), and the functor is $\mathrm{Mod}(\varphi)$ restricted to the elementary embeddings, which are the natural transformations between coherent functors on the Morleyisations. If it is an equivalence of categories, Theorem 6.1 gives $\mathsf{P}\varphi$ an equivalence, so the pretopos completions of the Morleyisations are equivalent, and the theories are Morita equivalent by [22, Thm. 5.4]. □

Theorem 2 of [7] obtains, under the weaker hypothesis that only the isomorphisms of models are required to be the Σ-isomorphisms, Morita equivalence "up to external auxiliaries"; the corollary shows that with elementary embeddings as morphisms the qualification and the additional definability condition of [7, Thm. 3] are unnecessary. The groupoid case is the subject of Question 8.2.

Meadows [18] defines a *Morita interpretation* of S in T as a Morita extension T^+ of T together with an interpretation $t\colon S \to T^+$, shows that it induces a functor $t^\dagger\colon \mathrm{mod}(T) \to \mathrm{mod}(S)$ on the categories of models and elementary embeddings, $t^\dagger(M) = t_*(M^+)$ with M^+ the canonical expansion of M to T^+ [18, Def. 39, Thm. 42], and calls T and S *Morita bi-interpretable* if there are Morita interpretations in both directions whose induced functors are quasi-inverse up to definable natural isomorphisms [18, Def. 43–44]. Whether Morita bi-interpretability implies Morita equivalence is left open in the tables of [18, §5, §5.1.1], which do not use conceptual completeness.

Corollary 6.4 (Morita interpretations)**.** *Let T, S be classical first-order theories, with elementary embeddings as morphisms of models. If T Morita interprets S via (T^+, t) and the induced functor $t^\dagger\colon \mathrm{mod}(T) \to \mathrm{mod}(S)$ is an equivalence of categories, then T and S are Morita equivalent. In particular Morita bi-interpretability implies Morita equivalence, the natural isomorphisms being allowed to be arbitrary; conversely Morita equivalence is 1-isomorphism in Meadows's 2-category $\mathbf{TH}_{\mathrm{Mor}}$* [18, Cor. 47]*, hence 1-equivalence, so the two notions coincide.*

Proof. Morleyise: $(T^+)^m$ is a Morita extension of T^m [22, proof of Thm. 5.4], so $\mathcal{C}_{T^m} \to \mathcal{C}_{(T^+)^m}$ induces an equivalence of pretopos completions [22, Cor. 4.9]; and $\mathrm{mod}(T) = \mathrm{Mod}(\mathcal{C}_{T^m})$, $\mathrm{mod}(T^+) = \mathrm{Mod}(\mathcal{C}_{(T^+)^m})$, natural transformations between coherent functors on a Morleyisation being the elementary embeddings. The reduct functor $r\colon \mathrm{mod}(T^+) \to \mathrm{mod}(T)$ is $\mathrm{Mod}(j)$

for the inclusion $j\colon \mathcal{C}_{T^m} \to \mathcal{C}_{(T^+)^m}$, and $\mathsf{P}j$ is an equivalence, so r is an equivalence by Theorem 6.1. The canonical expansion $e\colon M \mapsto M^+$ is a functor with $re = \mathrm{id}$ [18, Thm. 42], so e is a quasi-inverse of r and an equivalence. Now $t^\dagger = \mathrm{Mod}(t^m) \circ e$ with $\mathrm{Mod}(t^m)\colon \mathrm{Mod}(\mathcal{C}_{(T^+)^m}) \to \mathrm{Mod}(\mathcal{C}_{S^m})$; if $t^\dagger$ is an equivalence so is $\mathrm{Mod}(t^m)$, and Theorem 6.1 makes $\mathsf{P}t^m\colon \mathsf{P}\mathcal{C}_{S^m} \to \mathsf{P}\mathcal{C}_{(T^+)^m} \simeq \mathsf{P}\mathcal{C}_{T^m}$ an equivalence. Thus S^m and T^m are Morita equivalent coherent theories [22, Thm. 4.7], and S, T are Morita equivalent [22, Thm. 5.4]. Morita bi-interpretability provides such a $t^\dagger$ with a quasi-inverse, definable or not. □

The same argument does not apply to the next row of Meadows's table, *Morita-congruence*, where $s^\dagger t^\dagger(M) \cong M$ is only required object by object [18, Def. 44(3)]: the functor $t^\dagger$ is then essentially surjective and injective on isomorphism classes but need not be full or faithful, and conceptual completeness needs the morphisms (Remark 6.2, Question 8.2). The corollary is to be distinguished from the results relating Morita equivalence to bi-interpretability in the model-theoretic sense of [17, Def. 5.5], where each theory interprets the models of the other as definable quotient structures and the two composites are definably isomorphic to the identity. McEldowney [17, Prop. 5.11] proves, for theories each proving that some sort has two distinct elements, that Morita equivalence *implies* bi-interpretability, the hypothesis serving to code the coproduct sorts; he shows that the converse fails for his binary sorts (Remark 5.5) and leaves it open for finite signatures. Kamsma [9, Fact 8.6] states the two notions to be equivalent for positively disjoint theories, Morita equivalence being read there, through [22], as having the same classifying topos, that is as V_{P}. Corollary 6.4 goes in the direction opposite to McEldowney's, from Meadows's Morita bi-interpretability, whose interpretations are identity-preserving and whose quotients and coproducts are supplied by the Morita extensions, to Morita equivalence; it needs no two-element hypothesis, and its content is Theorem 6.1, the Makkai–Reyes theorem, applied to Meadows's notion.

7. Conceptual distance as an alternation depth

[10] define, for a class X of theories with an equivalence relation E (there: definitional equivalence) and a symmetric relation S (there: "differ by one relation symbol", $T \subseteq T'$ conservatively with one new symbol), the *step distance* $d(T, T')$ as the least number of S-steps in a path $T = T_0, \dots, T_m = T'$ each step of which is an E-step or an S-step, and ∞ if there is no path; with these choices it is the *conceptual distance* [10, Def. 4.4–4.5].

Proposition 7.1. *Let E and S be as above and let $E \vee S$ be the join in the lattice of equivalence relations on X, the transitive closure of $E \cup S \cup S^{-1}$. Then $d(T, T') < \infty$ iff $T\,(E \vee S)\,T'$, and $d(T, T')$ is the alternation depth of the pair: the least k such that T and T' are connected by a chain $E\,S\,E\,S\cdots$ with k letters S. In particular the step distance is bounded on X iff the join*

$E \vee S$ is reached by a uniformly bounded alternation, and Theorem 4.7 of [10] *(every $n \in \mathbb{N} \cup \{\infty\}$ is a conceptual distance) states that for conceptual distance some pairs are not related by $E \vee S$, and that on the related pairs the alternation depth is finite at each pair but not uniformly bounded.*

Proof. A path with k letters S is an alternating chain with k S-steps after merging consecutive E-steps (E is transitive) and inserting identity E-steps; the two descriptions of the join agree. □

This is the situation of [21, Thm. 2.3(iii)] for two closure operators: the join of the two classes is the class of the join iff the alternation is finite at every point, and finiteness at every point does not give a uniform bound, as in the three-dimensional template example there. Conceptual distance is the alternation depth for the pair (definitional equivalence, one-symbol extension), and the theories of [10, Thm. 4.7] with distance n are the analogue of the sets at which n rounds are needed.

8. Duality: which comparison levels are recovered from the models

Maruyama [16] asks when a duality between a syntactic side and a semantic side emerges, changes and breaks. The results above give an answer for the comparison of theories, in the following form. Say that a comparison level $W \in \mathrm{Same}(\mathbf{Coh})$ is *dualisable through* a 2-functor $M\colon \mathbf{Coh}^{op} \to \mathcal{K}$ if $W = V_M$, and that the duality *holds* at that level if moreover $W = V_R$ for a reflector R whose image is recovered from M (the localisation at W is reflective and the reflection is determined by the models).

Theorem 8.1. *The Morita sameness V_{P} is dualisable through the category of models, $V_{\mathsf{P}} = V_{\mathrm{Mod}}$ (Theorem 6.1), and through the ultracategory of models, $V_{\mathsf{P}} = V_{\mathrm{Mod}_{\mathrm{ult}}}$; in the second case the pretopos is recovered as the category of ultrafunctors $\mathrm{Mod}(\mathcal{C}) \to \mathbf{Set}$ (Makkai's strong conceptual completeness* [13], [12, Cor. 11]*), so the duality holds.*

Proof. The first statement is Theorem 6.1. For the second, $\mathrm{Mod}_{\mathrm{ult}}(f)$ is an ultrafunctor, its ultrastructure maps being invertible, so if its underlying functor is an equivalence a quasi-inverse carries the transported ultrastructure and $\mathrm{Mod}_{\mathrm{ult}}(f)$ is an equivalence of ultracategories; thus $V_{\mathrm{Mod}_{\mathrm{ult}}} = V_{\mathrm{Mod}}$, and $\mathsf{P}\mathcal{C} \simeq \mathrm{Fun}_{\mathrm{ult}}(\mathrm{Mod}(\mathcal{C}), \mathbf{Set})$ is [13]. □

The object-level categorical equivalence of [3] is the situation where the semantic side is kept but the morphisms connecting the syntactic objects are forgotten; the duality then "breaks" in the sense that equivalent model categories no longer come from an interpretation. Adding structure on the semantic side (ultraproducts, or the topology of Awodey–Forssell [2] on groupoids of models) restores it.

Question 8.2. Let $\mathrm{Mod}_{\mathrm{gpd}}(\mathcal{C})$ be the groupoid of models and isomorphisms, the core of $\mathrm{Mod}(\mathcal{C})$ without the topology of [2]. Is $V_{\mathrm{Mod}_{\mathrm{gpd}}} = V_{\mathsf{P}}$, i.e. is a coherent functor of pretoposes inducing an equivalence of groupoids of models an equivalence? For ω-categorical theories the automorphism group of the countable model determines the theory up to bi-interpretability (Ahlbrandt–Ziegler [1]); in general the topological groupoid of [2] is needed to recover the classifying topos, and Makkai's [14] studies which structure on the groupoid of models makes definability recoverable. Hudetz's Theorem 2 [7] is a statement at this level, and Meadows's Morita-congruence [18, Def. 44], which constrains only the objects, lies below it.

9. Open questions

(1) The sharpness of the bound four (Section 5).
(2) The lattice generated by the five samenesses V_{prod}, V_{sub}, V_{F}, V_{E} and renaming in Same of the category of theories with signatures (rather than of **Coh**, where the cartesian ones are trivial): which joins are reached in one round, and the position of definitional equivalence, bi-interpretability [5] and the intermediate notions of [18], whose remaining open arrows (strict bi-interpretability against definitional equivalence, iso-congruence against strict bi-interpretability, Problems 21 and 23 of [18]) are questions about this lattice.
(3) Translations between logics (double negation, Girard, Baaz) as closure operators on theories, after [16, §4.10], and their joins with the Morita completion.

Declarations

The author has no competing interests to declare.

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Daiichi Institute of Technology
Email address: r-shibusawa@daiichi-koudai.ac.jp